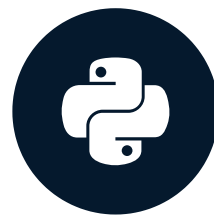


Summary statistics

DATA MANIPULATION WITH PANDAS



Maggie Matsui

Senior Content Developer at DataCamp

Summarizing numerical data

```
dogs["height_cm"].mean()
```

```
49.714285714285715
```

- `.median()` , `.mode()`
- `.min()` , `.max()`
- `.var()` , `.std()`
- `.sum()`
- `.quantile()`

Summarizing dates

Oldest dog:

```
dogs["date_of_birth"].min()
```

```
'2011-12-11'
```

Youngest dog:

```
dogs["date_of_birth"].max()
```

```
'2018-02-27'
```

The .agg() method

```
def pct30(column):  
    return column.quantile(0.3)
```

```
dogs["weight_kg"].agg(pct30)
```

```
22.599999999999998
```

Summaries on multiple columns

```
dogs[["weight_kg", "height_cm"]].agg(pct30)
```

```
weight_kg    22.6  
height_cm    45.4  
dtype: float64
```

Multiple summaries

```
def pct40(column):  
    return column.quantile(0.4)
```

```
dogs["weight_kg"].agg([pct30, pct40])
```

```
pct30    22.6  
pct40    24.0  
Name: weight_kg, dtype: float64
```

Cumulative sum

```
dogs["weight_kg"]
```

```
0    24
1    24
2    24
3    17
4    29
5     2
6    74
Name: weight_kg, dtype: int64
```

```
dogs["weight_kg"].cumsum()
```

```
0     24
1     48
2     72
3     89
4    118
5    120
6    194
Name: weight_kg, dtype: int64
```

Cumulative statistics

- `.cummax()`
- `.cummin()`
- `.cumprod()`

Walmart

```
sales.head()
```

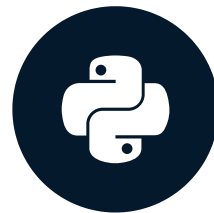
	store	type	dept	date	weekly_sales	is_holiday	temp_c	fuel_price	unemp
0	1	A	1	2010-02-05	24924.50	False	5.73	0.679	8.106
1	1	A	2	2010-02-05	50605.27	False	5.73	0.679	8.106
2	1	A	3	2010-02-05	13740.12	False	5.73	0.679	8.106
3	1	A	4	2010-02-05	39954.04	False	5.73	0.679	8.106
4	1	A	5	2010-02-05	32229.38	False	5.73	0.679	8.106

Let's practice!

DATA MANIPULATION WITH PANDAS

Counting

DATA MANIPULATION WITH PANDAS



Maggie Matsui

Senior Content Developer at DataCamp

Avoiding double counting



Vet visits

```
print(vet_visits)
```

```
   date      name      breed  weight_kg
0 2018-09-02  Bella  Labrador    24.87
1 2019-06-07    Max  Labrador    28.35
2 2018-01-17  Stella  Chihuahua     1.51
3 2019-10-19   Lucy  Chow Chow    24.07
..      ...      ...      ...      ...
71 2018-01-20  Stella  Chihuahua     2.83
72 2019-06-07    Max  Chow Chow    24.01
73 2018-08-20   Lucy  Chow Chow    24.40
74 2019-04-22    Max  Labrador    28.54
```

Dropping duplicate names

```
vet_visits.drop_duplicates(subset="name")
```

	date	name	breed	weight_kg
0	2018-09-02	Bella	Labrador	24.87
1	2019-06-07	Max	Chow Chow	24.01
2	2019-03-19	Charlie	Poodle	24.95
3	2018-01-17	Stella	Chihuahua	1.51
4	2019-10-19	Lucy	Chow Chow	24.07
7	2019-03-30	Cooper	Schnauzer	16.91
10	2019-01-04	Bernie	St. Bernard	74.98
(6	2019-06-07	Max	Labrador	28.35)

Dropping duplicate pairs

```
unique_dogs = vet_visits.drop_duplicates(subset=["name", "breed"])  
print(unique_dogs)
```

	date	name	breed	weight_kg
0	2018-09-02	Bella	Labrador	24.87
1	2019-03-13	Max	Chow Chow	24.13
2	2019-03-19	Charlie	Poodle	24.95
3	2018-01-17	Stella	Chihuahua	1.51
4	2019-10-19	Lucy	Chow Chow	24.07
6	2019-06-07	Max	Labrador	28.35
7	2019-03-30	Cooper	Schnauzer	16.91
10	2019-01-04	Bernie	St. Bernard	74.98

Easy as 1, 2, 3

```
unique_dogs["breed"].value_counts()
```

```
Labrador      2
Schnauzer     1
St. Bernard   1
Chow Chow     2
Poodle        1
Chihuahua     1
Name: breed, dtype: int64
```

```
unique_dogs["breed"].value_counts(sort=True)
```

```
Labrador      2
Chow Chow     2
Schnauzer     1
St. Bernard   1
Poodle        1
Chihuahua     1
Name: breed, dtype: int64
```

Proportions

```
unique_dogs["breed"].value_counts(normalize=True)
```

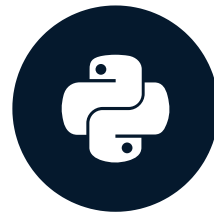
```
Labrador      0.250  
Chow Chow     0.250  
Schnauzer     0.125  
St. Bernard  0.125  
Poodle        0.125  
Chihuahua     0.125  
Name: breed, dtype: float64
```

Let's practice!

DATA MANIPULATION WITH PANDAS

Grouped summary statistics

DATA MANIPULATION WITH PANDAS



Maggie Matsui

Senior Content Developer at DataCamp

Summaries by group

```
dogs[dogs["color"] == "Black"]["weight_kg"].mean()  
dogs[dogs["color"] == "Brown"]["weight_kg"].mean()  
dogs[dogs["color"] == "White"]["weight_kg"].mean()  
dogs[dogs["color"] == "Gray"]["weight_kg"].mean()  
dogs[dogs["color"] == "Tan"]["weight_kg"].mean()
```

```
26.0  
24.0  
74.0  
17.0  
2.0
```

Grouped summaries

```
dogs.groupby("color")["weight_kg"].mean()
```

```
color
Black    26.5
Brown    24.0
Gray     17.0
Tan        2.0
White    74.0
Name: weight_kg, dtype: float64
```

Multiple grouped summaries

```
dogs.groupby("color")["weight_kg"].agg([min, max, sum])
```

	min	max	sum
color			
Black	24	29	53
Brown	24	24	48
Gray	17	17	17
Tan	2	2	2
White	74	74	74

Grouping by multiple variables

```
dogs.groupby(["color", "breed"])["weight_kg"].mean()
```

```
color  breed
Black  Chow Chow    25
       Labrador    29
       Poodle      24
Brown  Chow Chow    24
       Labrador    24
Gray   Schnauzer    17
Tan     Chihuahua     2
White  St. Bernard   74
Name: weight_kg, dtype: int64
```

Many groups, many summaries

```
dogs.groupby(["color", "breed"])[["weight_kg", "height_cm"]].mean()
```

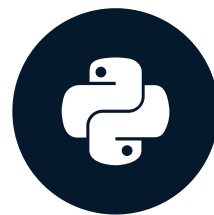
		weight_kg	height_cm
color	breed		
Black	Labrador	29	59
	Poodle	24	43
Brown	Chow Chow	24	46
	Labrador	24	56
Gray	Schnauzer	17	49
Tan	Chihuahua	2	18
White	St. Bernard	74	77

Let's practice!

DATA MANIPULATION WITH PANDAS

Pivot tables

DATA MANIPULATION WITH PANDAS



Maggie Matsui

Senior Content Developer at DataCamp

Group by to pivot table

```
dogs.groupby("color")["weight_kg"].mean()
```

```
color
Black    26
Brown    24
Gray     17
Tan        2
White    74
Name: weight_kg, dtype: int64
```

```
dogs.pivot_table(values="weight_kg",
                   index="color")
```

```
           weight_kg
color
Black             26.5
Brown             24.0
Gray              17.0
Tan                2.0
White             74.0
```

Different statistics

```
import numpy as np
dogs.pivot_table(values="weight_kg", index="color", aggfunc=np.median)
```

	weight_kg
color	
Black	26.5
Brown	24.0
Gray	17.0
Tan	2.0
White	74.0

Multiple statistics

```
dogs.pivot_table(values="weight_kg", index="color", aggfunc=[np.mean, np.median])
```

	mean	median
	weight_kg	weight_kg
color		
Black	26.5	26.5
Brown	24.0	24.0
Gray	17.0	17.0
Tan	2.0	2.0
White	74.0	74.0

Pivot on two variables

```
dogs.groupby(["color", "breed"])["weight_kg"].mean()
```

```
dogs.pivot_table(values="weight_kg", index="color", columns="breed")
```

breed	Chihuahua	Chow Chow	Labrador	Poodle	Schnauzer	St. Bernard
Black	NaN	NaN	29.0	24.0	NaN	NaN
Brown	NaN	24.0	24.0	NaN	NaN	NaN
Gray	NaN	NaN	NaN	NaN	17.0	NaN
Tan	2.0	NaN	NaN	NaN	NaN	NaN
White	NaN	NaN	NaN	NaN	NaN	74.0

Filling missing values in pivot tables

```
dogs.pivot_table(values="weight_kg", index="color", columns="breed", fill_value=0)
```

breed	Chihuahua	Chow Chow	Labrador	Poodle	Schnauzer	St. Bernard
color						
Black	0	0	29	24	0	0
Brown	0	24	24	0	0	0
Gray	0	0	0	0	17	0
Tan	2	0	0	0	0	0
White	0	0	0	0	0	74

Summing with pivot tables

```
dogs.pivot_table(values="weight_kg", index="color", columns="breed",  
                  fill_value=0, margins=True)
```

breed	Chihuahua	Chow Chow	Labrador	Poodle	Schnauzer	St. Bernard	All
color							
Black	0	0	29	24	0	0	26.500000
Brown	0	24	24	0	0	0	24.000000
Gray	0	0	0	0	17	0	17.000000
Tan	2	0	0	0	0	0	2.000000
White	0	0	0	0	0	74	74.000000
All	2	24	26	24	17	74	27.714286

Let's practice!

DATA MANIPULATION WITH PANDAS